Project Preliminary Report

Matthew Abel

1. Introduction

Within this project I intend to explore the relationship within horse betting markets between race number and market accuracy. For a given horse racing event, there will be a number of races throughout the day (typically 7 or now in some cases 8). It has been told to me as somewhat of a gamblers' folk tale that prices tend to be less accurate as the day continues, as gamblers move towards riskier options in hopes of recouping earlier losses. This behaviour would imply that betting on the favourite for the final race would give you a greater expected value than betting on the favourite for the first race.

I have identified 2 primary datasets that I can use for this research. Both are provided by popular betting exchange service Betfair, where users are able to back and lay bets against other users on the platform. Betfair provides a large dataset through their historic data program which I have been using already to begin this research, however I have come across another dataset provided by them which is likely more appropriate moving forward. The dataset I have been using features an enormous amount of datapoints including price movements and closely follows the internal API structure used to place bets. The other dataset is more slimmed down, providing only information collated after the races have occured. Importantly too, the other dataset has data from a much greater time period and contains information (such as market volume) which is only available at a cost for the original dataset. The most important part of both these datasets is the start price, which is an aggregation of all bet demand when the race starts.

Other sources that may be useful are publications with predictions such as Racing Post or Timeform. Often these sources don't just publish which horses they believe are most likely to win, but also what they believe that the betting odds will be for them on the racecourse.

To look into this potential factor, I will apply multiple different metrics of accuracy. Probably most important is the Brier score, which measures the accuracy of odds compared to the real outcome. Similarly, log-loss can be used to evaluate the effectiveness of the market in producing accurate prices. Another factor to look at would be how the favourite-longshot bias changes from race to race. This is an observation in how punters typically prefer to bet on outsiders rather than the favourite. All of these factors can be cross-compared with other important data points such as the liquidity, earlier prices, price ranges, race length and going. I can use the Kruskal-Wallis test to determine if the differences per race are statistically significant. I think it may also be worth comparing some basic betting strategies and seeing how they perform.

2. Background Material

There is not much (if any) existing research into this particular feature that I can find. However, there is a significant amount of other research into horse racing markets.

The favourite-longshot phenomena in which unfavourable horses are often highly overweighted was first published by Griffith (1949). This has been greatly analysed from a psychological perspective, with many arguing that this is a preference for risky bets among gamblers (Weitzman, 1965; Ali, 1977; Asch, 1987; Kanto, 1992). Others, such as Golec and Tamarkin (1998) or Snowberg and Wolfers (2010) suggest different reasons including a preference for skew rather than risk or simply a misperception of actual probabilities. None of these studies however examine how the favourite-longshot bias evolves over the course of a race day.

An important area of relevant research is that of market efficiency. Work in this area provides useful groundwork to model what market efficiency looks like in betting markets and using these methods will enable me to compare market efficiency (or price accuracy). There is relevant research from Snyder (1978), Hausch et al. (1981), Asch and Quandt (1987), and Busche and Walls (2000) just to name a few to start with.

3. Schedule

Tasks must be completed by:

- May 22 - Data acquisition
- Jul 3 - Software development
- Jul 10 - Software testing
- Jul 31 - Written analysis & evaluation
- Aug 3 - Final deadline

References

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